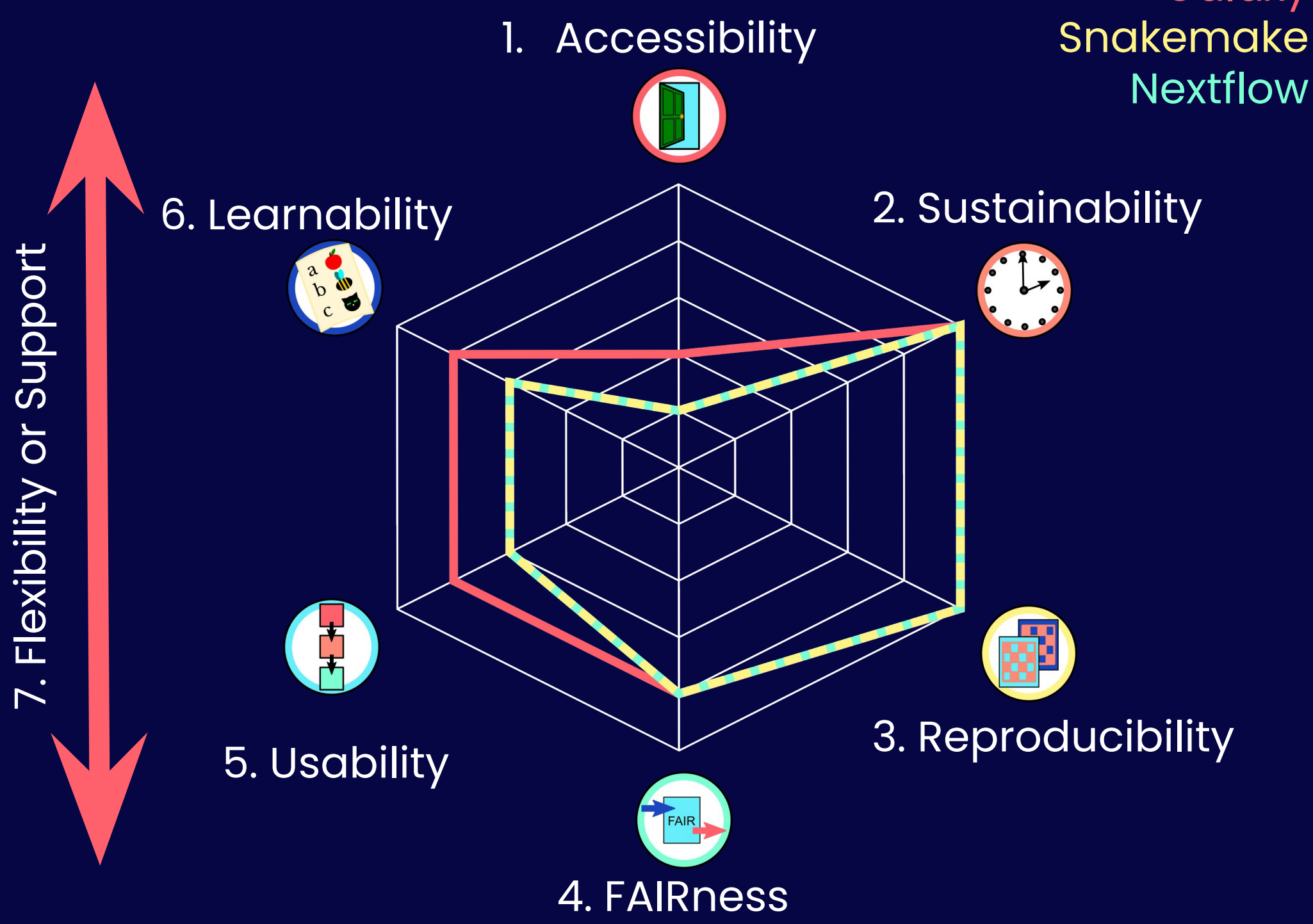
Why Use Galaxy?

Comparing the options for user-friendly bioinformatics

Aim & Methods: A scoping review was performed with the aim of identifying the features of bioinformatics Workflow Management Systems (WMS) that enhance user experience. Accessibility, reproducibility, and sustainability were among the key user needs repeatedly identified in the literature. Usability evaluations were performed on each platform based on the characteristics identified during the review. The preliminary results are presented here.

7 Characteristics of WMS



Conclusions & Future Work: Seven key characteristics of bioinformatics platforms were identified that are important for learners, researchers, and other users. The criteria used to score platforms during this review will be used to create guidelines for future evaluations by users, educators, or developers who are choosing or improving these platforms.

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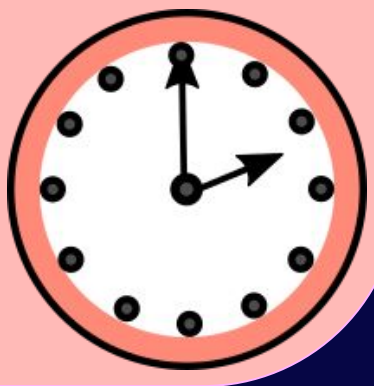
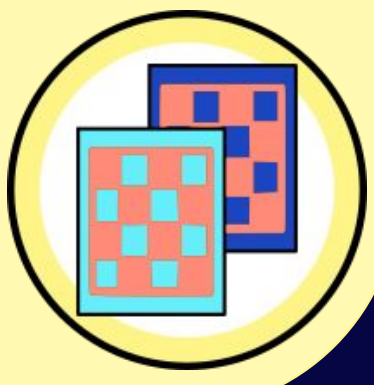
1. Accessibility: a clearly and consistently laid out interface that provides alternative modes of viewing and interacting with content

2. Sustainability: an established and well-maintained platform with an active community of users / contributors

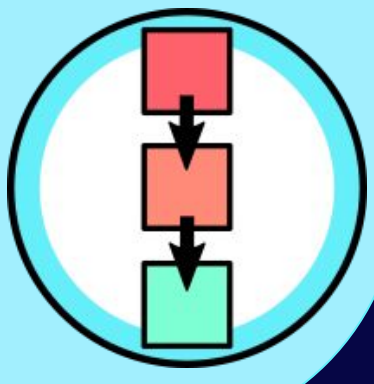


3. Reproducibility: automatically captures all the information needed to reproduce analyses, workflows can easily be rerun, exported, imported, and adapted



4. FAIRness: outputs conform to FAIR data standards - they are Findable, Accessible, Interoperable, and Reusable

5. Usability: platform is easy to install, understand, navigate and use, so workflows can be created, edited, and run efficiently - with clear information provided on progress and errors



6. Learnability: sufficient guidance is provided to use unfamiliar tools, with comprehensive training materials available to explain the WMS and the analyses it runs

7. Flexibility-Support: platforms may give more control to skilled users, enabling greater flexibility, or provide simpler interfaces that reduce the risk of errors and offer more support to the user

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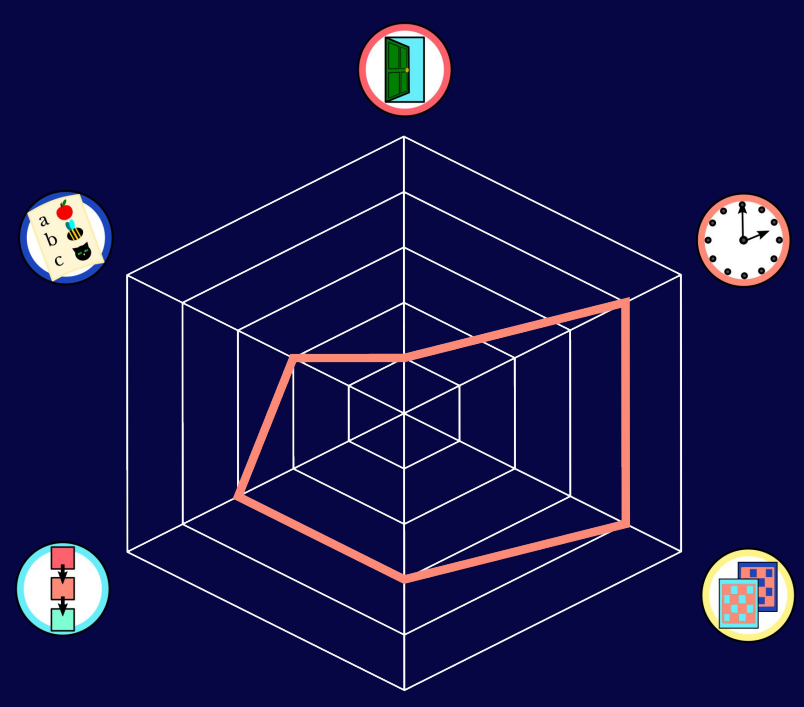
We're here to help you

UGENE

Galaxy

Nextflow

Snakemake



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You take charge

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